Lab5

Marker (測距離) & Tello EDU

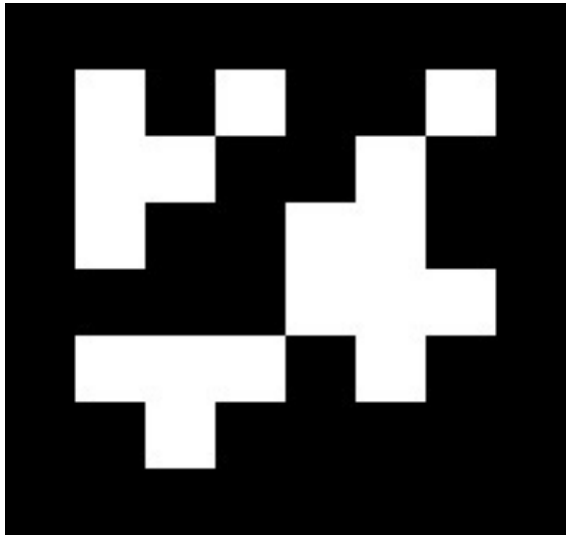
安裝套件

- pip install opevcv-contrib-python==**4.4.0.46**
- pip install djitellopy

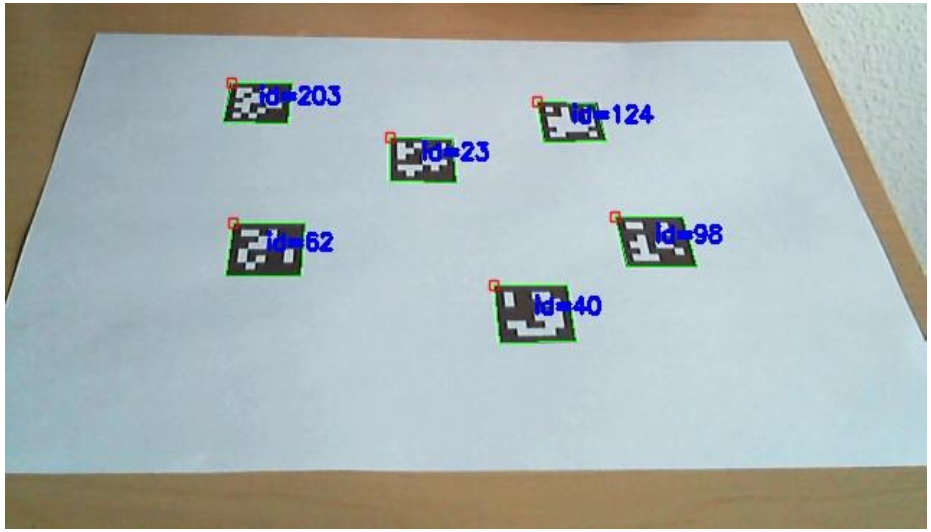
1. Marker Detection (50%)

- Calibration
- **Marker Detection**
- **Pose Estimation**
- Controlling

1. Marker Detection (50%)



aruco marker



marker detection

1. Marker Detection (50%)

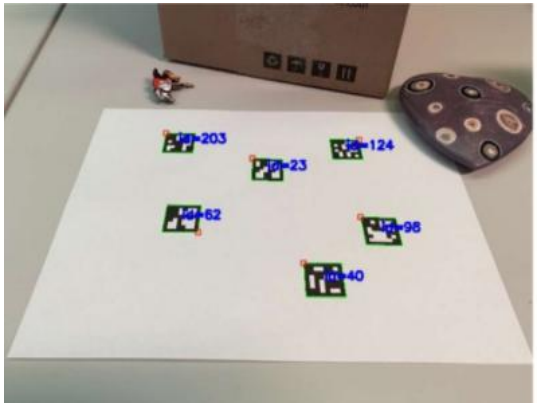
```
# Load the predefined dictionary
dictionary = cv2.aruco.Dictionary_get(cv2.aruco.DICT_6X6_250)

# Initialize the detector parameters using default values
parameters =  cv2.aruco.DetectorParameters_create()
```

1. Marker Detection (50%)

```
# Detect the markers in the image
markerCorners, markerIds, rejectedCandidates = \
cv2.aruco.detectMarkers(frame, dictionary, parameters=parameters)

frame = cv2.aruco.drawDetectedMarkers(frame, markerCorners, markerIds)
```



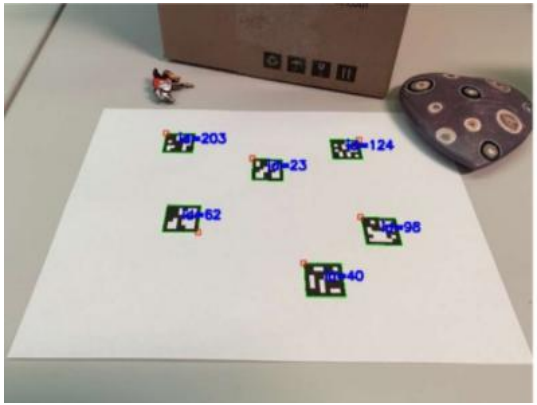
1. Marker Detection (50%)

15: marker size

```python
#Pose estimation for single markers.
rvec, tvec, _objPoints = \
cv2.aruco.estimatePoseSingleMarkers(markerCorners, 15, intrinsic, distortion)

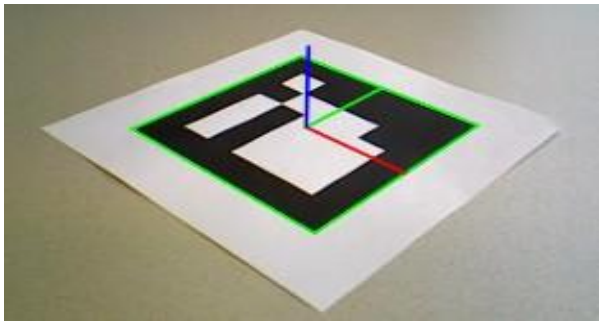
frame = cv2.aruco.drawAxis(frame, intrinsic, distortion, rvec, tvec, 0.1)
```

```python
# Get the parameters of camera calibration
fs = cv2.FileStorage("calibrateCamera.xml", cv2.FILE_STORAGE_READ)
intrinsic = fs.getNode("intrinsic").mat()
distortion = fs.getNode('distortion').mat()
```



1. Marker Detection (50%)

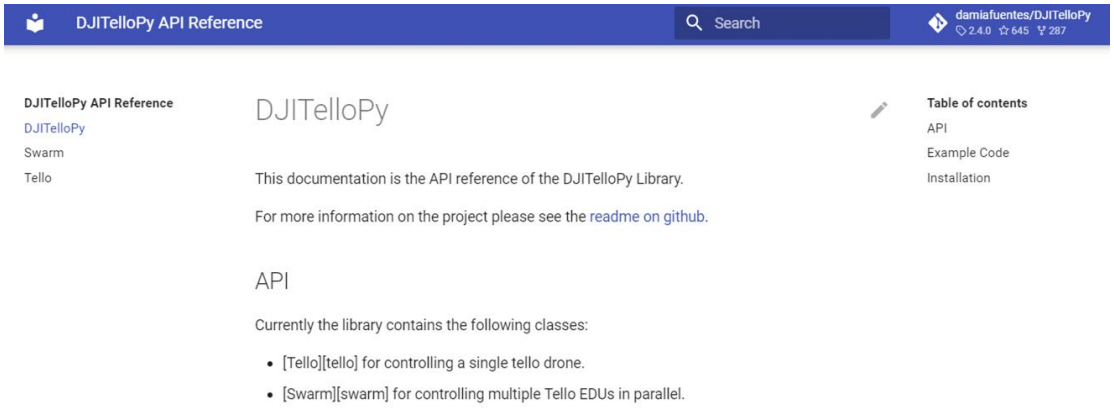
- Calibrate the drone camera
- Marker detection by drone camera
- Pose estimation



x: 10.3478
y: 21.5618
z: 3.9908

2. Tello EDU (50%)

- djitellopy



[DJITelloPy API Reference]

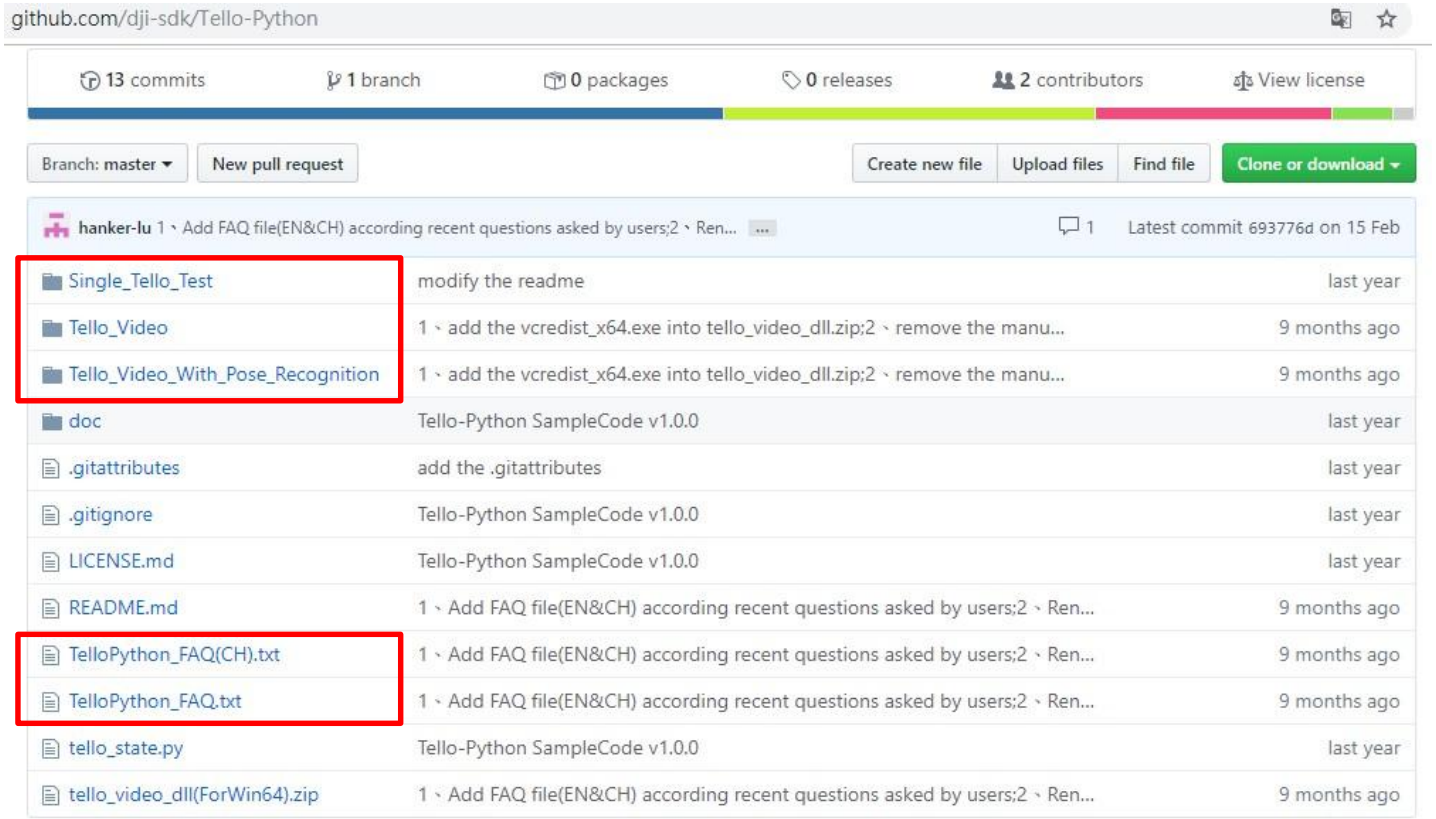
2. Tello EDU (50%)

配件		數量
	飛行器	× 1
	螺旋槳（對）	× 4
	槳葉保護罩（套）	× 1
	電池	× 1
	Micro USB 傳輸線	× 1
	螺旋槳拆卸工具	× 1
	挑戰卡	× 4



2. Tello EDU (50%)

- https://github.com/dji-sdk/Tello-Python



2. Tello EDU (50%)

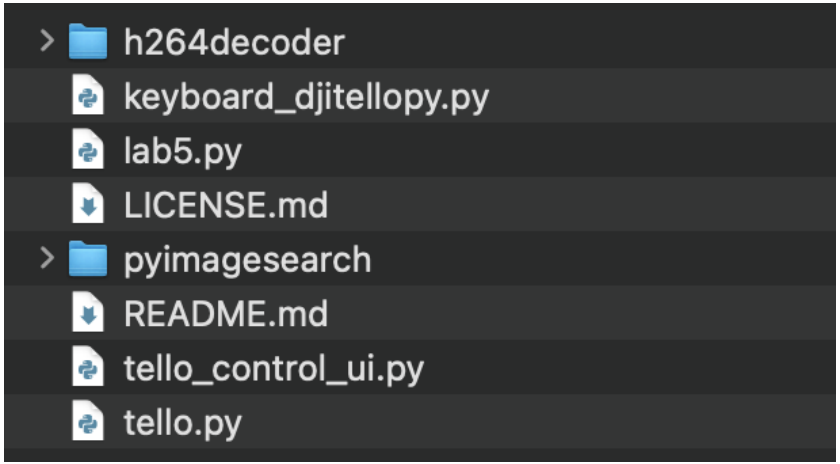
- 打開Tello無人機，並透過Wi-Fi將筆電連接到Tello





2. Tello EDU (50%)

- 取像控制
  - 在 Tello_Video 資料夾內放入 lab5.py




```python
import cv2
import numpy as np
import time
import math
from djitellopy import Tello
from pyimagesearch.pid import PID

def main():
    # Tello
    drone = Tello()
    drone.connect()
    #time.sleep(10)
    drone.streamon()
    frame_read = drone.get_frame_read()


    while True:
        frame = frame_read.frame

        cv2.imshow("drone", frame)
        key = cv2.waitKey(33)

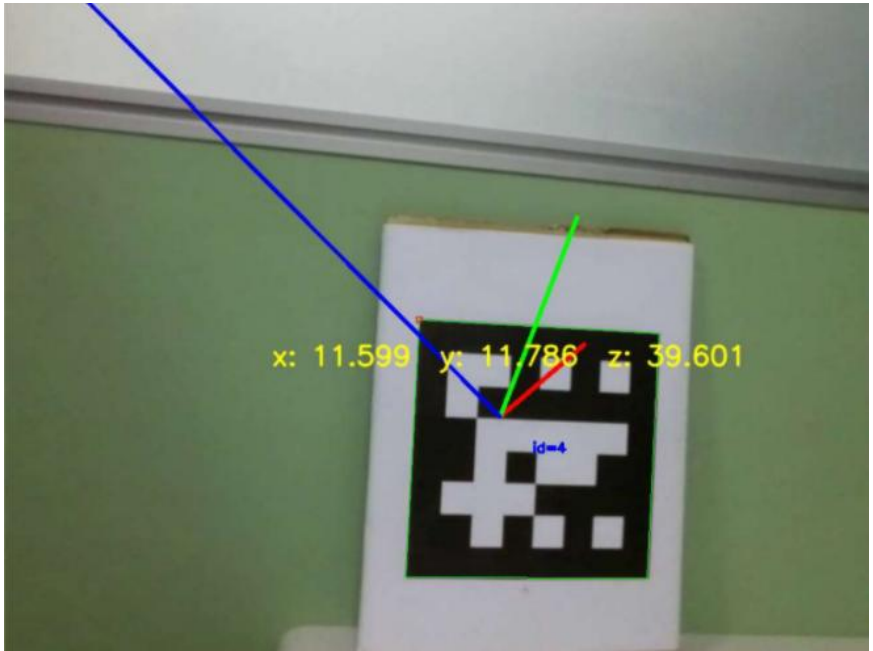
    #cv2.destroyAllWindows()



if __name__ == '__main__':
    main()
```

2. Tello EDU (50%)

- 測距離 - **cv2.putText()** 把位置放上去
  - 除了x, y, z, aruco id，請印出真實的距離
  - marker size = 15 cm



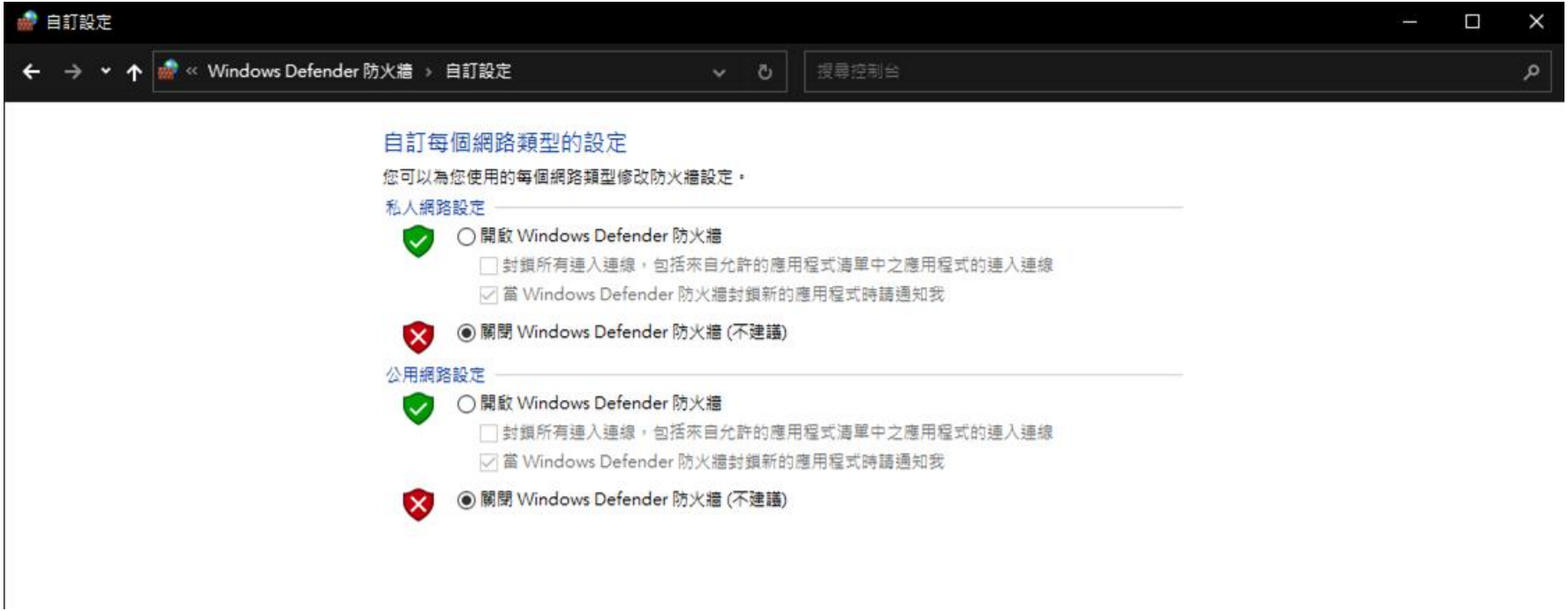
2. Tello EDU (50%)

- 若找不到 aruco module
  - pip install opencv-contrib-python

```
(cvdronelab) C:\Users\raymo>pip list |findstr opencv
opencv-contrib-python 4.4.0.46
opencv-python         4.4.0.46
```

2. Tello EDU (50%)

- 若無人機無法取像，嘗試關閉防火牆 (飛完記得打開)



注意

本週不要讓無人機起飛!!!!

起飛的那組就當作這週作業未完成